

Physics Informed Neural Networks for the transient diffusivity equation

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THE 1D CARTESIAN PDE

The 1D cartesian PDE, for the diffusivity equation is expressed and defined as follows;

$$\frac{\partial}{\partial x} \left(K_x \frac{\partial P}{\partial x} \right) = \phi \mu c_t \frac{\partial P}{\partial t} \quad (1)$$

$$\frac{\partial^2 P}{\partial x^2} k_x = \phi \mu c_t \frac{\partial P}{\partial t} \quad (2)$$

Making $\frac{\partial^2 P}{\partial x^2}$ stand alone, we have

$$\frac{\partial^2 P}{\partial x^2} = \frac{\phi \mu c_t}{K_x} \frac{\partial P}{\partial t} \quad (3)$$

Derivation of the dimensionless PDE for equation scaling in PINNs

Using (3) and transforming it into it's dimensionless form, we have;

- Dimensionless space $\tilde{x} = \frac{x}{x_e}$
- Dimensionless time $\tilde{t} = \frac{t}{t_{max}}$

Evaluating the LHS of equation (3), we have;

$$\frac{\partial^2 P}{\partial x^2} = \frac{1}{x_e^2} \frac{\partial^2 P}{\partial \tilde{x}^2} \quad (4)$$

Evaluating the RHS of equation (3), we have;

$$\frac{\partial P}{\partial t} = \frac{1}{t_{max}} \frac{\partial P}{\partial \tilde{t}} \quad (5)$$

Equating equations (4) and (5), we have;

$$\frac{\partial^2 P}{\partial \tilde{x}^2} = \frac{\phi \mu c_t x_e^2}{k_x t_{max}} \frac{\partial P}{\partial \tilde{t}} \quad (6)$$

$$\text{Since } t_{D,max} = \frac{k_x t_{max}}{\phi \mu c_t x_e^2}$$

$$\frac{\partial^2 \tilde{P}}{\partial \tilde{x}^2} = \frac{1}{t_{D,max}} \frac{\partial \tilde{P}}{\partial \tilde{t}} \quad (7)$$

Derivation of the PDE used to solve the simultaneous inverse problem

I derived the inverse PDE that was used to solve the permeability and distance away from the sealing boundary simultaneously instead of solving them sequentially to reduce the training time from two to one, the trainable variables η_k and η_l are defined below.

From Equation (7), Multiply both sides by $\frac{1}{t_{D,max}}$, we have;

$$\frac{\partial \tilde{P}}{\partial \tilde{t}} = \frac{\partial^2 \tilde{P}}{\partial \tilde{x}^2} t_{D,max}$$

$$\text{Since } \eta_k = \frac{k_x}{k_{max}} \text{ and } \eta_l = \frac{x_e}{x_{e,max}}$$

$$t_{D,max} = \frac{k_x t_{max}}{\phi \mu c_t x_e^2}$$

$$\frac{\partial \tilde{P}}{\partial \tilde{t}} = \frac{\partial^2 P}{\partial \tilde{x}^2} \frac{\eta_k}{\eta_l^2} \left(\frac{k_{max} t_{max}}{\phi \mu c_t x_{e,max}^2} \right) \quad (8)$$

Derivation of the flowrate equation used to solve the inverse simultaneous problem

I derived the flowrate equation used to solve for the simultaneous inverse problem, which are as follows;

$$\tilde{q} = \frac{kA}{\mu} \frac{d\tilde{p}}{dx} \quad (9)$$

Solving for $\frac{d\tilde{p}}{dx}$ then obtain $\frac{\partial\tilde{p}}{\partial x}$ via auto diff we have;

$$d\tilde{p} = (p_i - p_{wf})\partial\tilde{p}$$

$$dx = (x_e)\partial x$$

$$\frac{d\tilde{p}}{dx} = \left(\frac{p_i - p_{wf}}{x_e} \right) \frac{\partial\tilde{p}}{\partial x} \quad (10)$$

$$\tilde{q} = \frac{kA}{\mu} \frac{d\tilde{p}}{dx} \quad (11)$$

$$d\tilde{p} = (p_i - p_{wf})\partial\tilde{p}$$

$$dx = (x_e)\partial x$$

$$q_{calc} = \frac{\eta_k k_{max_inverse} A}{\mu} \left(\frac{p_i - p_{wf}}{\eta_l x_{max_inverse}} \right) \left(\frac{\partial\tilde{p}}{\partial x} \right) \quad (12)$$

Setting up the Initial and Boundary conditions

Section 1: The initial condition is defined as follows;

$$p(x, t = 0) = p_o = 25.0[MPa]$$

For case 1.1: The well is controlled by a constant BHP at well ($x_w = 0$) and a no-flow boundary at x_e at 500m.

$$p(x_w = 0, t) = p_{wf} = 3.0[MPa]$$

$$x_e \left. \frac{\partial p}{\partial x} \right|_{x_e} = 0$$

Result obtained for case 1.1(Forward problem)

The Neural Network was able to satisfy the initial and boundary conditions, it was able to learn the PDE from equation (7) to produce the desired output. The no-flow boundary is 500m away from the well.

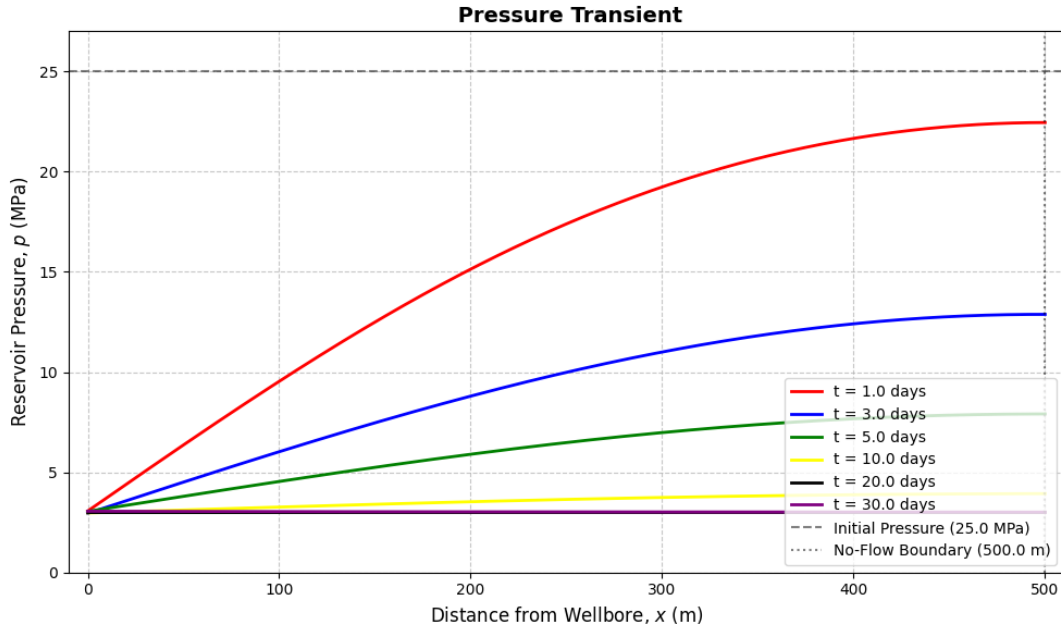


Figure 1: One no-flow boundary to the right

Section 1.2: The initial condition is defined as follows;

$$p(x, t = 0) = p_o = 25.0[MPa]$$

case 1.2: The well is placed inside the domain $x_w \in [-500, 750]$ at well placed at $x_w = 0$. Two no-flow boundaries at $x_{e1} = -500$ and $x_{e2} = 750$ m

$$p(x_w = 0, t) = p_{wf} = 3.0[MPa]$$

$$x_{e1} \left. \frac{\partial p}{\partial x} \right|_{x_{e1}} = x_{e2} \left. \frac{\partial p}{\partial x} \right|_{x_{e2}} = 0$$

Result obtained for case 1.2 (Forward problem)

The well was placed in-between two no-flow boundaries, this problem was solved using domain decomposition by decomposing the neural network into two subdomains, training subdomain 1 to the left and subdomain 2 to the right then connecting both subdomains by a common interface which is the well. The Neural Network was able to satisfy the initial and boundary conditions, it was able to learn the PDE from equation (7) to produce the desired output.

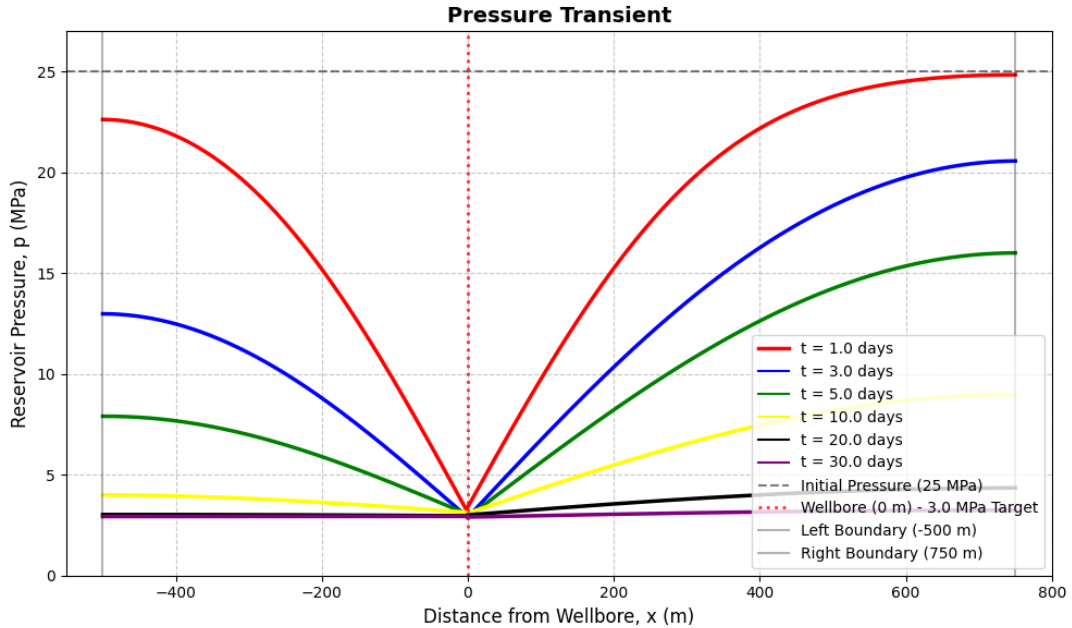


Figure 2: Two no-flow boundary to the left and right

Obtaining the flowrate history

After the output (pressure) has been obtained from the forward model, the flowrate is to be calculated using equation (9), the gradient $\frac{d\bar{p}}{dx}$ is calculated via automatic differentiation by applying the exact chain rule of calculus backwards through the computational graph from the output (pressure) to the input (distance).

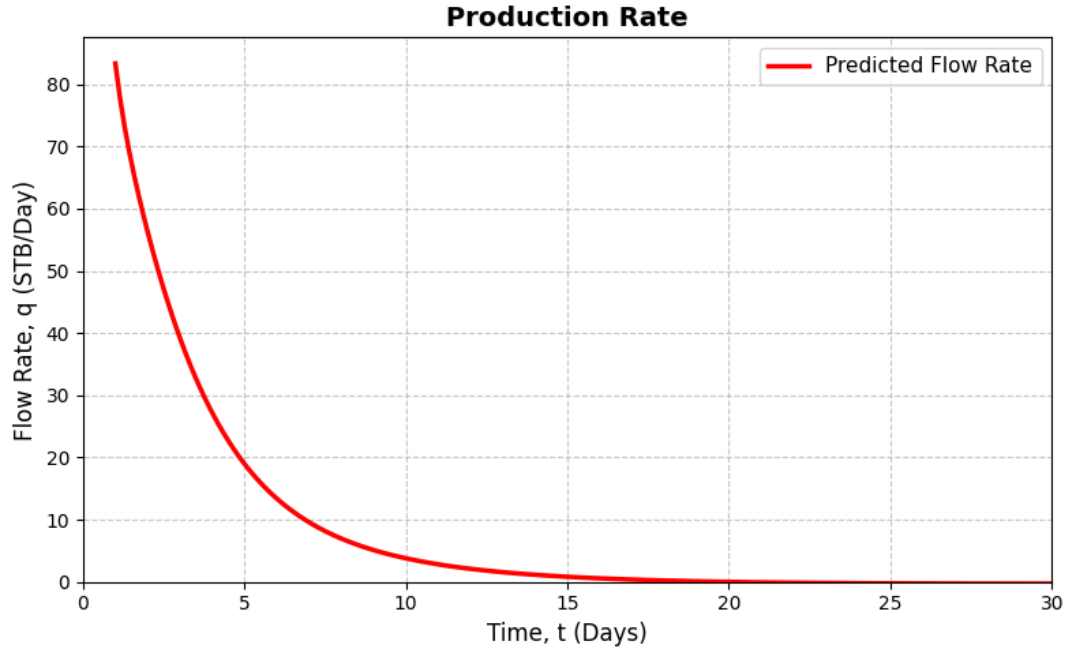


Figure 3: The flowrate history prediction

Setting up the inverse problem

The PINN model should be able to learn the unknown trainable permeability and unknown trainable boundary distance values from equation (8) which are η_k and η_l that satisfies that PDE, Initial and Boundary conditions and the injected production history data.

For me to verify that the PINN successfully learnt the PDE and was able to obey the physics of the PDE, that PINN should be able to obtain parameters somewhat accurately from the PDE itself. I took the predicted flowrate obtained in figure (3) and added noise to it.

The addition of this noise would cause a little difference in the permeability and boundary distance values prediction used in the forward model which was 120md and 500m respectively. The noisy flowrate which is going to be injected into the inverse problem model, where a predicted flowrate is also going to be obtained and it's going to be solved via gradient descent using equation (12) above.

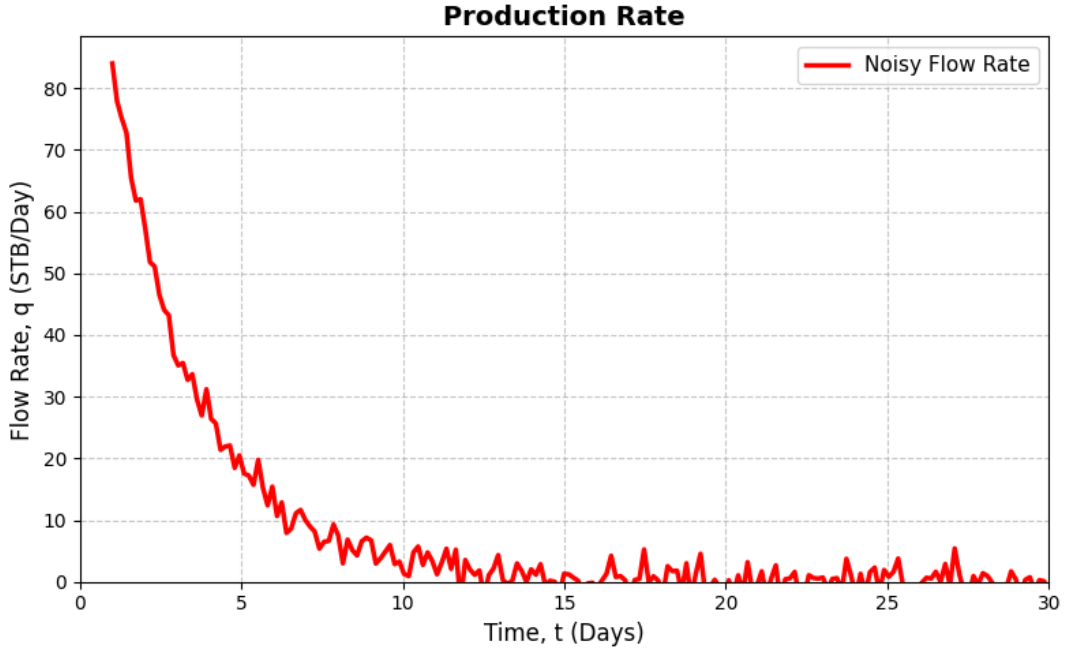


Figure 4: The noisy flowrate history prediction

The inverse problem predictions

The permeability and boundary distance away from the wellbore predictions are 125.90md and 499.14m respectively for the first case. We can see that the PINNs was able to learn the PDE efficiently and was able to give sound results after training for 30,000 epochs.

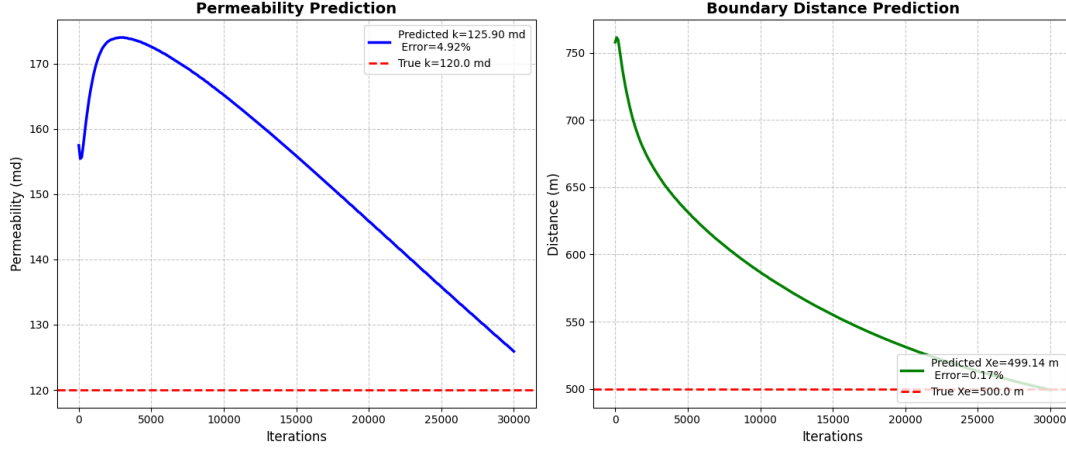


Figure 5: Permeability and boundary distance predictions: First run

After running the PINN model for a second time, I found that a change in the inverse problem parameters is due to the inherent stochasticity built into how neural networks operate. Running the inverse PINN model multiple times would help calculate an uncertainty bounds to gain enough confidence in the model predictions.

The second model run gave permeability predictions as 116.58 md and boundary distance as 498.03m.

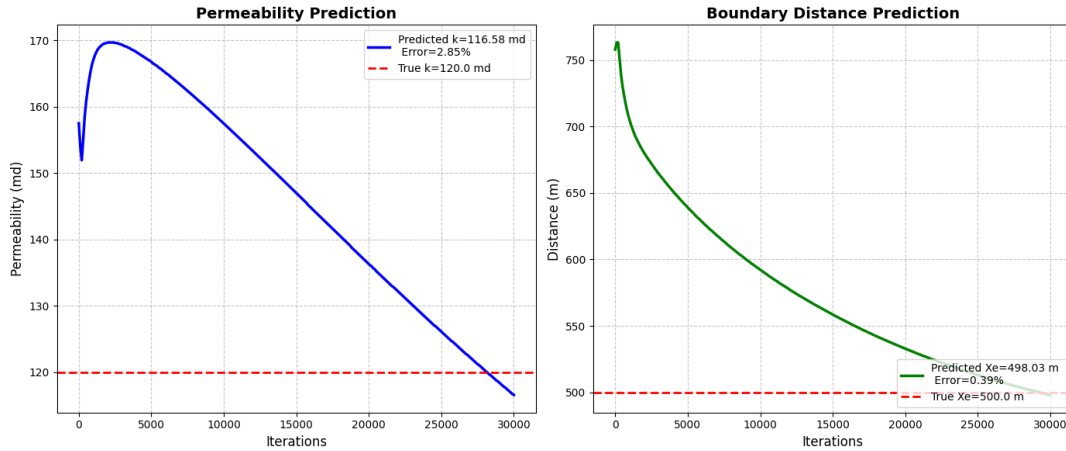


Figure 6: Permeability and boundary distance predictions: Second run

The cylindrical Radial PDE

The cylindrical radial PDE is expressed and derived as follows;

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k_r r \frac{\partial p}{\partial r} \right) = \phi \mu c_t \frac{\partial p}{\partial t} \quad (13)$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) = \frac{\phi \mu c_t}{k_r} \frac{\partial p}{\partial t} \quad (14)$$

Evaluating the LHS of equation (14), we have;

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) = \frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} \quad (15)$$

The cylindrical Radial PDE is then defined as;

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} = \frac{\phi \mu c_t}{k_r} \frac{\partial p}{\partial t} \quad (16)$$

Derivation of the dimensionless PDE for equation scaling in PINNs

Using (16) and transforming it into its dimensionless form, we have;

- Dimensionless space $\tilde{r} = \frac{r}{r_e}$
- Dimensionless time $\tilde{t} = \frac{t}{t_{max}}$

Evaluating the RHS and LHS of equation (16);

$$\frac{1}{t_{D,max}} \frac{\partial \bar{P}}{\partial \tilde{t}} = \frac{1}{\tilde{r}} \frac{\partial \bar{P}}{\partial \tilde{r}} + \frac{\partial^2 \bar{P}}{\partial \tilde{r}^2} \quad (17)$$

Setting up the Initial and Boundary conditions

Section 2: The initial condition is defined as follows;

$$p(x, t = 0) = p_o = 25.0[MPa]$$

Both cases for section 2, the outer boundary is a no-flow boundary at $q_e = 0$

$$r_e \left. \frac{\partial p}{\partial r} \right|_{r_e} = 0$$

For case 2.1: The well is producing at a constant BHP

$$p(x_w = 0, t) = p_{wf} = 3.0[MPa]$$

For case 2.2: The well is producing at a constant rate

$$r_w \left. \frac{\partial p}{\partial r} \right|_{r_w} = \frac{q_w \mu}{2\pi k h} = p_{ch}$$

Where p_{ch} is the characteristics pressure

Result obtained for case 2.1(Forward problem)

This forward problem for the cylindrical radial PDE was solved by utilizing Domain decomposition primarily Conservative PINNs (CPINNs). This method utilizes the law of conservation of mass to effectively divide a cell into subdomains to effectively model a nonlinear 2D PDE, an interface condition is also defined between subdomains to connect those multiple subdomains to model the PDE effectively in such cases.

Since we are modeling the pressure transient for the cylindrical case, it is imperative that the pressure continuity and pressure fluxes(derivatives) from multiple subdomains are equal. The interface condition is placed at 50m away from the wellbore to model the violent pressure drop at that wellbore due to the small cross sectional area.

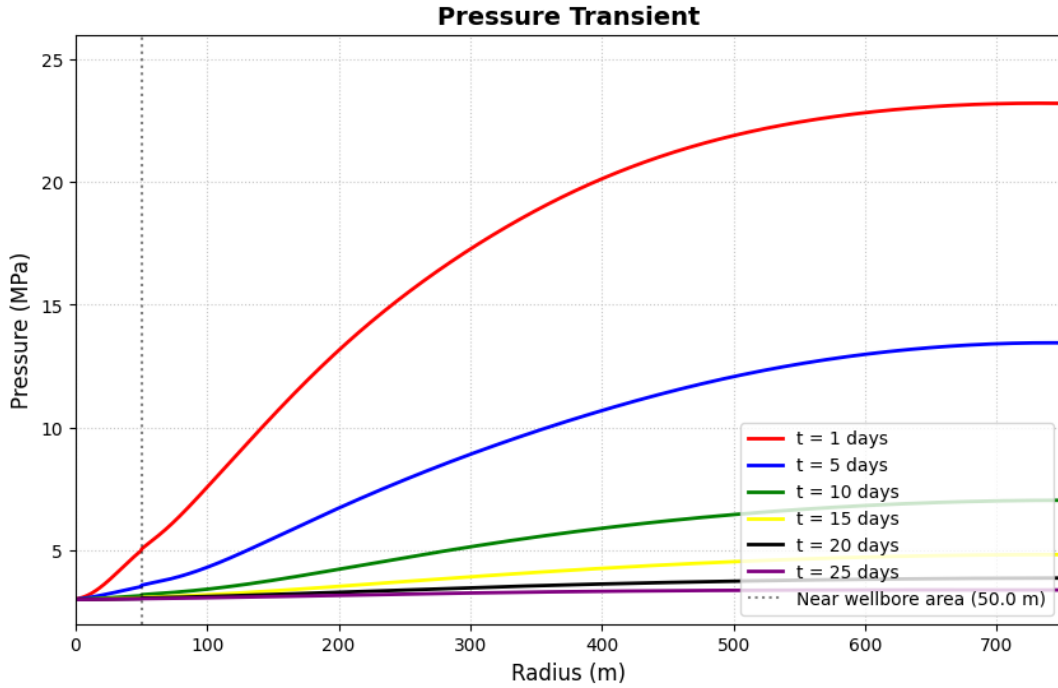


Figure 7: Pressure transient for the cylindrical 2D case

Obtaining the flowrate history

After the output (pressure) has been obtained from the forward model, the flowrate is to be calculated using equation (9), the gradient $\frac{d\bar{p}}{dx}$ is calculated via automatic differentiation by applying the exact chain rule of calculus backwards through the computational graph from the output (pressure) to the input (distance).

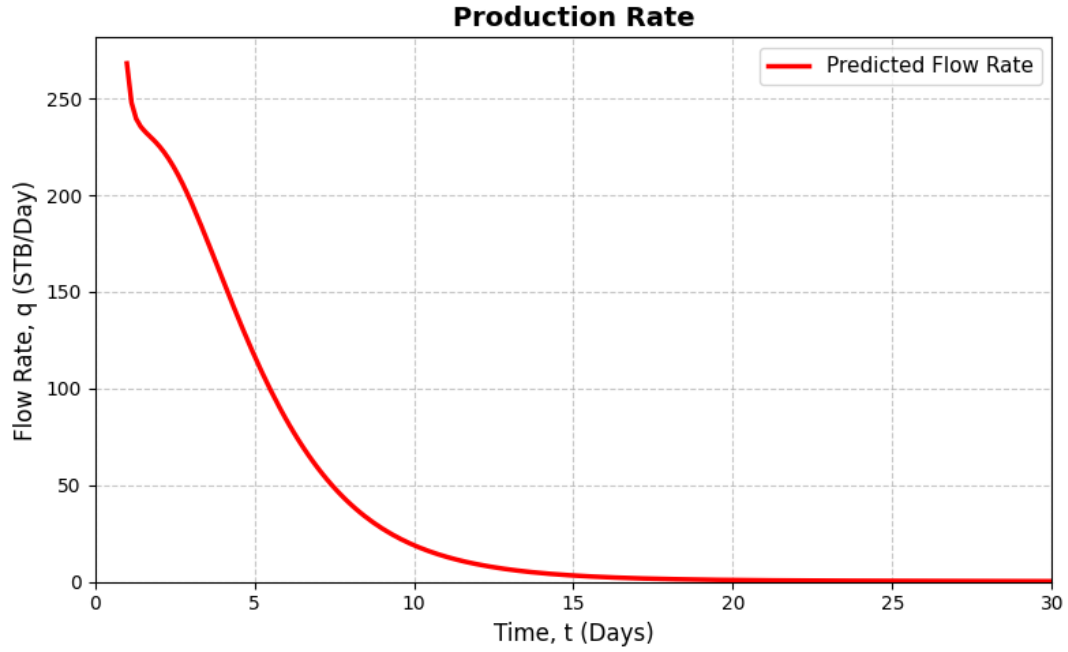


Figure 8: Flow rate prediction from the forward cylindrical model

Results obtained for case 2.2 (forward model)

In this case, we are dealing with a well producing at a constant rate, the initial, well and boundary conditions are noted above. This problem was also solved by utilizing domain decomposition of the radial cell to effectively model the violent pressure drop at the wellbore, looking at the image below, we can see the violent pressure drops at days 60 and 80 which is caused by the small cross sectional area at the near wellbore region. I was able to model the near wellbore region and farther region in the reservoir using subdomains, both subdomains were connected in the radial cell at an interface of 50m as seen below where the pressure continuity and pressure fluxes constraints are checked due to the law of conservation as Conservative PINNs was used in this case.

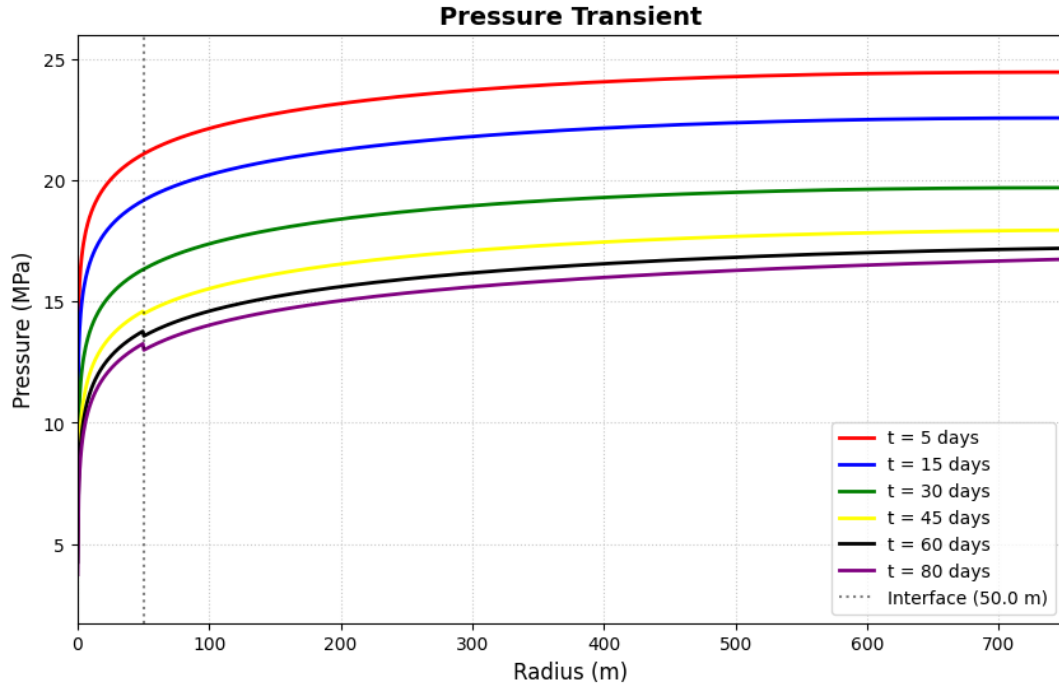


Figure 9: Pressure transient for the constant rate case